

Commercial LEO Destination Phase 2 Certification and Services Industry Acquisition Update

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Industry Briefing Ground Rules

- ❖ Any industry questions shall be submitted in writing via SAM.GOV
 - Responses will be uploaded to SAM.GOV at a later date
- ❖ The Government will not respond to questions or comments at this briefing
- ❖ This chart package will be posted on SAM.GOV

Baseline Acquisition Strategy

- ❖ NASA intended to purchase end-to-end mission services from the commercial owner/operator of the CLD, including:
 - Crew pre-flight and on-orbit training
 - Pre- and post-flight payload processing
 - Crew and cargo transport to and from the CLD
 - In-flight accommodation and support for crew, payloads, activities, studies, and technology demonstrations, etc.

- ❖ The goal of this approach was to maximize CLD partner involvement and their ability to support NASA's and their other customers' needs while NASA concentrates on ensuring crew safety and NASA mission objectives are completed
 - NASA insight and oversight included joint mission authority for the safety of NASA crew and the execution of NASA research

Baseline Acquisition Strategy

- ❖ Long term goal for NASA is to become one of many customers for commercial transportation and operations in LEO, purchasing services from commercial providers to meet the agency's needs
 - NASA will require a minimum of two NASA astronauts continuously living in LEO by 2031
 - CLD capabilities will support the LEO Microgravity Strategy Goals & Objectives
 - Decomposition of the goals and objectives to characteristics & needs, functions and use cases will be done Jan-Feb, 2025 and synced to CLDP draft requirements
- ❖ A full and open competition for services will be conducted among U.S. commercial providers.
 - CLD providers may propose an evolutionary approach to develop destination capabilities
 - On-ramp opportunities will be included for future commercial advancements
- ❖ An overlap between CLD and ISS capabilities is desired with need for CLD services by 2029
- ❖ CLD providers will develop, own and operate their destinations, with NASA support as requested through RSAAs or GTAs
- ❖ CLD and transportation providers will be responsible for meeting NASA's certification requirements

Challenges of Transition



**No Gap in
Low Earth
Orbit**



**Timeline
Flexibility**



**Continuity of
Payload and
Transportation
Capability**



**Market
Maturity**



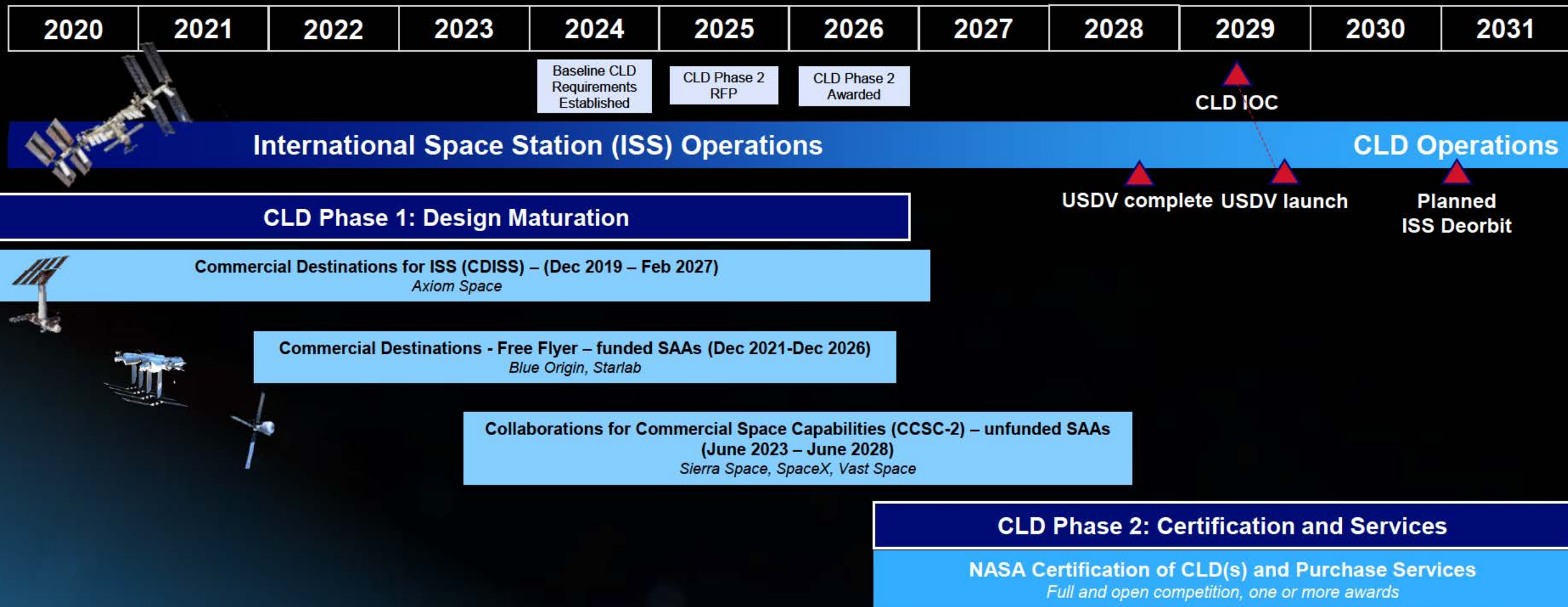
Acquisition Strategy Updates

The following updates to the strategy are being implemented to address NASA identified concerns:

- ❖ Initially, NASA will provide transportation for all missions associated with NASA services (Crew and Cargo)
 - NASA will provide transportation for an initial set of missions, by contracting directly with transportation providers, and transition to CLD end-to-end service (including transportation) over time (TBD)
- ❖ CLD providers may propose an evolutionary approach with the following additional assumptions:
 - Minimum Initial Operating Capability (Min IOC) required by December 2029 (Details of Min IOC still in work)
 - Short-term presence including crew tended of 2 NASA crew for 1-3 months would be acceptable.
 - Full capability required by December 2031
 - Includes 2 NASA crew continuously in LEO for 6 month missions
 - Additional shorter or longer missions may also be of NASA interest
- ❖ CLD providers will develop, own and operate their destinations, with NASA support as requested through RSAAs or GTAs. NASA looking to offer more support/services than previously planned:
 - Open action to finalize list of services/equipment to be offered
 - Includes potentially offering versus requiring (TBC) NASA provided Exploration subsystems



LEO Transition Schedule





CLD Acquisition Strategy Schedule (Notional)



- **Detailed Acquisition schedule is under development based on ASM updates and alignment with LMS decomposition schedule**
 - Overall Acquisition schedule has not changed
 - Potential for acceleration of schedule also being considered
 - * LMCR is the LEO Microgravity Strategy Concept Review
- **Additional Industry touch points are also planned prior to draft RFP**
 - Virtual Industry Day to discuss Acquisition plans/answer questions
 - Requirements and Safety Workshop in January with release of documents
 - Additional workshops/TIMs are being planned to solicit feedback on areas such as Legal, liability, Minimum IOC, Minimum Buy, and Evaluation criteria



Summary of Acquisition Approach Decisions

- ❖ ASM Approval to proceed with Acquisition Strategy Option that included:
 - CLD & Transportation contracted for separately initially to address end to end service complexities and stability of the transportation industry
 - Increased GTA/GFE/RSAA offerings to assist CLD providers decrease schedule and technical risk where needed
 - Use of Evolutionary Approach with Minimum IOC that support LEO Microgravity Strategy Objectives and decrease risk of gap
 - NASA may accept offerings of short duration mission/crew tended, if proposed from 2029-2031, to allow for early services during overlap period with ISS to decrease risk of a gap in LEO